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A Comprehensive Review of IoT-based Health Monitoring Systems

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Abstract—**In the rapid emergence and evolution of technology, individuals often resemble working machines, neglecting a crucial aspect - health. This review explores IoT-based health monitoring systems, emphasizing the shortcomings and features of existing solutions. IoT is a new internet ecosystem evolving rapidly in research, especially in healthcare. This research study presents the implementation of an IoT-based remote patient monitoring system, where authorized persons can access and monitor patients in remote locations. The analysis includes a detailed examination of Quality of Service (QoS) challenges and trends. This study proposes an IoT-based conceptual architecture for monitoring cardiac and chronic diseases by employing devices like heartbeat sensors, blood saturation sensors, ECG sensors, and blood pressure sensors. Coupled with microcontrollers, data is collected and transmitted to a local server. This system, available via an Android application, enables the virtual sharing of data with both physicians and users, allowing for the remote management of drug infusion, marking a substantial advancement in promoting health awareness.**

Keywords— Internet of Things (IoT), Health Monitoring Systems, Heart Beat Sensors, Blood Saturation Sensors, Electrocardiogram (ECG) Sensors, Blood Pressure Sensors, Microcontrollers, Wearable Biomedical Devices, Wireless Sensor Area Network (WSN), Medication Delivery System.

I. Introduction

In today's rapidly changing world, it's crucial to focus on meeting the growing health needs and concerns. Recent global health crises, like the pandemic, highlight the importance of healthcare. The search involves exploring how cutting-edge technology can be utilized to address current challenges within the diverse range of systems present in the market. This review explores IoT (Internet of Things) based health monitoring systems, analysing their shortcomings and noteworthy features. IoT emerges as a transformative force in healthcare, offering innovative solutions to enhance patient monitoring and promote well-being. The intersection of IoT and healthcare is particularly exemplified in the proposed IoT-based remote patient monitoring system, aiming for enhanced individualized care through the continuous monitoring of crucial indicators such as ECG, heart rate, and blood pressure [1].

The technological framework of these systems is complex and integrates a range of sensors such as DHT11, BMP180, heart beat module, and ECG, among others, coupled with diverse hardware components such as Arduino Uno, Nano, and Raspberry Pi 3. Complemented by services such as GSM modules and Node MCU, these systems collect data in numeric form, storing it in microprocessors. Every piece of data is gathered in numerical form and kept in microprocessors. Numerous apps are available to transfer data from these microprocessors to Cloud Fire storage, hosting, storing, and granting access to Android apps. All of these features require an account, which is protected by Google Security. The data is kept in an up-to-date database and retrieved using an Android application developed with Google's App Inventor, a tool that was previously provided by MIT. Admin, patients, and doctors are the recipients of this app. The patient and doctor can only read the data; only the administrator is authorized to write and read it. [2] The IoT-based system is a comprehensive approach to health monitoring, addressing issues ranging from cardiac conditions to chronic diseases. The utilization of sensors like body temperature, accelerometer, blood pressure, pulse detector, breathing sensor, glucose sensor, and electrochemical sensor forms a sensor network that provides an overview of an individual's health. The cost-efficient and adaptable Raspberry Pi is essential in the functionality of this health monitoring system.

The Internet of Things (IoT) plays a crucial role in the health monitoring system by enabling real-time data collection, transmission, and analysis. IoT facilitates continuous monitoring of patients' vital signs and provides remote access to healthcare professionals, thus enhancing patient care and reducing response times in emergencies. Also discussed about the IoT-based model termed the "Smart Portable Intensive Care Unit (SPICU)" which encloses this transformative potential offering remote diagnostic capabilities. In scenarios like the transfer of an ambulance, SPICU guarantees the remote availability of a physician, mitigating potential risks linked to delays in reaching medical facilities. The evolution witnessed in computer networks and data communications have now extended to object-to-object and object-to-computer communications, marked by the era of IoT. [3] This transformation is significantly impacting the health domain, enhancing the efficiency of monitoring processes. The SPICU model serves as an example, as it not only transmits real-time vital parameters but also enables the live transmission of video for remote diagnosis. The comprehensive database keeps track of everything from patient data to diagnostic decisions.

With developments taking place in several industries, wireless technology has become more and more popular in recent years. The latest Internet of Things buzz has truly caught up with key industries like the industrial sector, especially in recent years. Automation and control have been the main areas of experience with this insight. Another modern trend in high-quality healthcare is biomedical. IoT technology has created opportunities for both personal healthcare facilities and hospitals. Because it is an intelligent

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system, it monitors several parameters that use energy, cost money, and boost productivity.

In the initial phase, health checks are performed by physicians. It will take a reasonable amount of time to register, consult, and observe in this practice. Later on, reports were also produced. People in the workforce are reluctant or put off watching since it takes time. This rapid approach cuts down on the amount of time the mission may require. Important intermediaries in the domains of innovation and research have been advancing health care and human well-being for a long time. It is impossible to take away from or undercut their contribution to medicine. The modern vehicle systems bring with them the straightforward guidelines of the past and the early identification of chronic illnesses [1].

The most important factors, such as blood pressure, breathing rate, heart rate, and body temperature, will be used to gauge how serious the illness is. The communication channel that is subsequently connected to the server through Wi-Fi is Bluetooth. The system consists of a remote server, a smartphone, and a portable terminal. One of the most important aspects of staying healthy is regulating body temperature, particularly for the aged or disabled. This is because if the body temperature rises for one reason or another, that person may undergo a stroke. Additionally, it can be augmented at any moment of the human body temperature. So, the embedded computer is going to continuously monitor their temperatures and pressures and bring forward prompt alerts to the Caretaker. IoT and wireless sensing technology need to be used for effective control. Every thirty seconds, the web server gathers and displays the data from the sensor technology. It is too set to signal a warning to a doctor in cases where measurement data crosses the thresholds. Besides, a switch would be mounted on the hand of the patient to stand by for help in case the patient needs some help.

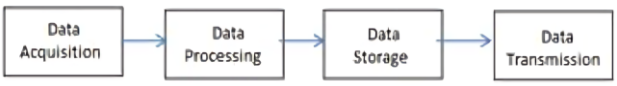

Fig. 1. IOT Application stages

A comprehensive IoT application's operation stages are: a. data acquisition; b. data processing; c. data storage; and d. data transmission. While processing and storage may or may not be present in some applications, the initial and end stages are present in all applications. In this case, real-time raw data transmission and real-time on-board processes are transmitted via data acquisition. The MEMS technology will be of great importance in reducing data acquisition energy. The compressed sensing paradigm can be exploited in many IoT applications for use because they have the data sparsity property.

We cover the compressed sensing technology in depth in the following sections, along with how wireless body sensor networks and health monitoring apps make use of it. To do this, there are two essential methods: There are two ways to improve the energy efficiency of the processing unit: 1) Extremely low power processors; and 2) Co-processors that are efficiently customized. By integrating an ultra-low-power near-threshold CPU with a high-performance processor and employing a job scheduling framework, energy efficiency is introduced to Internet of Things applications. Even though the field of embedded system design has conducted a great deal of research on memory energy reduction, there are some characteristics unique to Internet of Things applications that can be taken advantage of to further improve the energy efficiency of memory in IoT embedded devices. Moreover, radio transceivers that are integrated into it can help realize this SoC for better data transmission and low-power multi-radio chips. To minimize the quantity of data that needs to be communicated or stored, new compression techniques should be used to the data streams. [21]

This becomes a landmark study in healthcare technology of IoT-based systems. The seamless integration of sensors, microprocessors, and cloud-based platforms with a system beyond the just traditional bounds of healthcare ensures that the patient is well taken care of and that his health management is done proactively. This paper seeks to contribute to the continued dialogue on how the Internet of Things can be utilized in the betterment of healthcare and hence propelling the realization of a connected, informed, and healthy society.

II. Literature Review

The National Accreditation Board for Testing and Calibration Laboratories (NABL) and the Ministry of Science and Technology have introduced the Smart Portable Intensive Care Unit (SPICU) system, which uses Internet of Things (IoT) technology to function as a drug delivery and health monitoring system. A medical device with five parameters—heart rate, non-invasive body temperature, blood pressure, blood saturation, and electrocardiogram—is affixed to the patient in order to administer medication. The patient's vital signs are transmitted in JSON (JavaScript Object Notation) format to a cloud-based server. A cloud-based database is employed to access and update patient-related information via the creation of desktop and mobile applications. This allows physicians to observe a patient's real-time physiological parameters through a mobile application that also has the capability to remotely manage the infusion pump. [3] The mobile application retrieves patient data and enables the communication of doctor's instructions with a cloud-based server. A notification is also delivered to the doctor's cell phone when a patient is in the ambulance. Using Java Base Web Tokens (JWT) for authentication, the doctor can use the mobile application to view the patient's important parameters after gaining access. Notably, SPICU incorporates a crucial drug delivery feature, allowing doctors to command drug administration through the connected infusion pump inside the ambulance via IV routes.

In order to address the problem of time wastage during ambulatory services, Naina Gupta et al. proposed a structure. However, in hospitals, they were compelled to transmit the data using a GSM module linked via Bluetooth technology. routine physical examinations and the use of sensors attached to the body to monitor a variety of bodily characteristics. They concentrated on creating a compact wearable device that can send data to customized networks using GPRS [4].

A prototype developed by M. Ryan Fajar Nurdin et al. is limited to using Zigbee wireless technology for the transmission of electrocardiographic signals. They created a web application to display the corresponding readings based on the patients' data after affixing several sensors to various

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body nodes and gathering a series of analog data. Their program is only available to 20 people, and the data has various bandwidth problems that skew the results [5].

Melisa Pereira et al. used body fat percentage and heart rate as their main components in an effort to address the problem of wireless data transfer. They employed an Arm 7 processor, which operates on Wi-Fi data transfer, for the suggested approach. Still, the system as a whole is limited by the use of technology that is not appropriate for long-distance broad communication [6].

The goal of the method developed by Tati Erlina et al. is to monitor the physiological state of comatose patients by keeping a watch on their heart rate, respiration rate, and state of their eyelids. The goal of the complete arrangement is to transfer the data to an Android app that is downloaded and installed on a smartphone, displaying different measured readings on the app. Nevertheless, the solution exposed the sensitive data and was unable to maintain the security of the data within the system [7].

An Internet of Things (IoT)-based fitness tracking system for senior citizens was proposed by Ashwini Gutte et al. Numerous indicators, including blood pressure, temperature, heart rate, body glucose levels, ECG, and activity monitoring, are integrated into the system. The Raspberry Pi is used in the implementation. In order to monitor heartbeat and other conditions, Gatte additionally used an ECG (Electrocardiogram) sensor. Gatte also suggested various IoT-based protocols that may be used to put this kind of system into practice [8].

Niket Patil et al. developed an Internet of Things-based system for soldiers to monitor and be apprehended. Patil suggested employing sensors for monitoring, such as the oxygen spotter detection system, pulse rate sensor, and LM35. They used the GPS SIM28M to track their exact location. The Node MCU ESP 8266 Wi-Fi module connects the entire arrangement to the internet, and also has a panic button for emergency help. The Arduino Uno (AT Mega 328p), the MCU board, is coupled to sensors in this low-cost system [9].

Kumar et al. monitor the patient's body temperature, heart rate, movement, and respiration rate using a Raspberry Pi and sensors. Using infrared transmitters and receivers, heartbeat rate can be observed. This technique uses general circuitry or simple electronics to increase signal gain because the amplifier used to transport the signal from range may attenuate it. The 230 V input voltage is reduced using a step-down transformer, which also converts it to 5 and 9 volts. In addition, a Switched Mode Power Supply (SMPS) is used to meet the various input needs of various sensors [10].

A system with a GSM module connected to a Raspberry Pi in an assembly of several health monitoring sensors was proposed by Chowdhary et al. This method was designed to convey health data directly to doctors over the phone in the form of direct messaging. Their goal was to reduce the workload for patient monitoring in hospitals by making it portable. They were able to store the data in the system and refresh the updated value at a rate of 20 seconds each reading [11].

Nubenthan et al. created a WBAN system with a wireless interface. To gather data from several sensors, they divided up the hardware and used Wi-Mon software. With illnesses like dengue, this data provides information on how serious a patient's condition is. Wi-Mon software's exact capabilities enable the collection of fundamental data for health monitoring, such as body temperature, blood pressure, ECG, oxygen saturation, and pulse rate. [12]

Khan and colleagues developed a system utilizing a mix of Raspberry Pi and Arduino Uno. They utilized an Arduino to gather all the data from the sensors, and a Raspberry Pi was used to record a video. They established a reading center that was updated every two minutes using a local server. The purpose of this approach was to address the concerns of older populations and patients who do not see their doctor on a regular basis. [13]

A real-time tracking system for intensive care units (ICUs) was introduced by Uddin et al. This system gathers data and sends it to a specific application by using body sensors interfaced with an Arduino Uno. This software aids in the monitoring of multiple parameters within a given range and connectivity. Utilizing IoT Cloud and IoT protocols is crucial because they offer the built application a variety of possibilities for data transmission. [14]

An Internet of Things-based architecture was presented by Zia Uddin Ahmed et al. to use wearable biomedical equipment to track patient health. The system uses sensors that are connected to a microcontroller and a GSM module to provide the user or their family with emergency information. The patient's body is directly connected to a large number of hardware devices, and users can obtain information from various locations thanks to dispersed data. [15]

The healthcare sector poses a significant global challenge for researchers. The system allows continuous remote monitoring of patient's vital signs and health metrics in real time. This can help detect emergencies early. Provides physicians/doctors access to patient's health data anytime from anywhere for review and diagnosis. Empowers patients to independently monitor their health and effectively manage chronic conditions with timely data and insights. [18] Systems are often designed to be low-cost, portable and comfortable for patients and elderly individuals to track health from home. Collecting large datasets over time through IoT-enabled devices allows for predictive healthcare using techniques like machine learning.

Healthcare apps enable adult and patient independence. The IoT sensors are of the presence of which used at the time for diagnoses and revaluations of their wellness state and transmitting alerts in illegal situations. The IoT device itself will instruct the patient accordingly if other minor issues are discovered. More accurate detection is enabled by the use of a continuous record of the health parameters of the body. People focus more on early disease detection and prevention these days. Furthermore, the newest generation of mobile phone technologies and their offerings have a significant influence on the advancement of various network types, such as 3G, Bluetooth, wireless LAN, and GSM.[21]

The integration of IoT (Internet of Things) technology in healthcare can offer numerous advantages through the utilization of sensors, actuators, and smart devices. Wireless Sensor Area Network (WSN) technology, in particular, facilitates the sensing, monitoring, and tracking of data through intelligent sensors. They can be categorized into two broad classes of health apps when developed for an IoT: single step and mixed mode.

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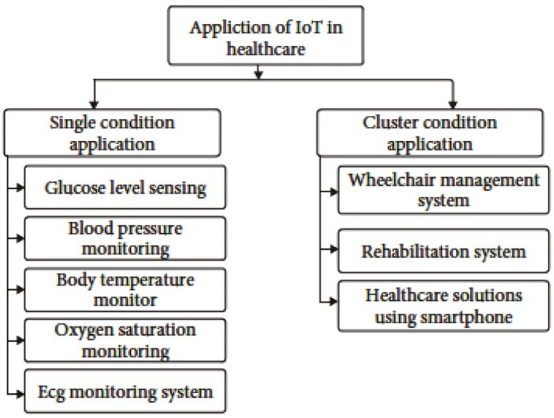

Fig. 2. Classification of applications in IoT based Healthcare [22].

Single status applications: These applications are intended for a single disease.

A. Glucose Sensor:

If a person has an above-average glucose level for an extended period of time, they may have diabetes. The blood glucose control system facilitates the formulation of blood glucose and aids in the prescription of appropriate tests, diet balances, and pharmaceuticals. A not allowed glucose-based m-IoT configuration process is now suggested here. Towards this purpose, the various sensors in the patients are interfaced together through the correct vendor of IPv6 connectivity. It designs within the OS a communication-based IoT device that propagates the data aggregated up to the sensor regarding blood glucose levels. This kit comprises a glucose gatherer, a workstation or smartphone, and the chipset. [22]

B. Blood Pressure Sensor:

The Oscillometric approach is utilized by the blood pressure sensor to determine the human body's systolic and diastolic blood pressure. This method involves detecting pressure fluctuations within an inflatable cuff wrapped around the upper arm. While the cuff undergoes inflation and deflation, the sensor gauges the magnitude and frequency of pressure fluctuations resulting from the rhythmic blood flow in the arteries. By analysing these oscillations, the sensor accurately determines both systolic (maximum) and diastolic (minimum) blood pressure values. This non-invasive technique provides a convenient and reliable means of monitoring blood pressure levels, assisting in the identification and control of hypertension and other cardiovascular disorders. [22]

C. Temperature Sensor:

The temperature range that the temperature sensor (LM35) is intended to measure is 55°C to 150°C. This sensor is ideal for tasks like monitoring body temperature because its output voltage shows a linear correlation with temperature, rising by 10 millivolts for each degree Celsius increase. Therefore, for every 1°C increase in body heat, the sensor's output voltage increases by 10 millivolts. This linear relationship simplifies temperature measurement and allows for precise monitoring of changes in body temperature over the specified range. [22]

D. Pulse Oximeter Sensor:

The pulse oximeter sensor (MAX30100) employs red and infrared lights to estimate oxygen saturation and heart rate by passing these lights through a finger, toe, nostril, or earlobe, allowing the sensor to measure changes in light absorption. By analysing the differences in absorption between oxygenated and deoxygenated blood, the sensor calculates the percentage of oxygen saturation in the body. Additionally, the sensor detects variations in blood volume caused by the heartbeat, enabling accurate measurement of heart rate. This non-invasive method provides real-time monitoring of oxygen levels and heart rate, making it valuable for medical applications and fitness tracking. [16]

E. ECG Sensor:

The AD8232 ECG sensor is specifically made to identify and quantify the electrical impulses produced by the heart muscle. However, these signals can often be very small and susceptible to interference from sources like electromagnetic interference (EMI) and movement artefacts. To ensure accurate and clear detection of the ECG signal, an op-amp (operational amplifier) is utilized. The op-amp amplifies the weak ECG signal while simultaneously filtering out unwanted noise and interference, resulting in a clean and reliable electrical signal. This capability enables precise analysis and monitoring of cardiac activity, enabling healthcare professionals to diagnose and treat cardiac conditions effectively. [22]

Fig. 3. Different types of Sensors

Consolidated status requests: These programs work together to treat specific illnesses.

A. Wheelchair Management System:

Professionals recommend the use of comfortable wheelchairs in saving elderly lives and the lives of those with disabilities. IoT has highly facilitated this process fast in this area. Numerous sensors are built into smart wheelchairs to track the position of the seat and provide information on the patient's or user's status. [22] [23]

B. Rehabilitation System:

The Internet of Things helps rehabilitate this system to regenerate problems related to the growth of the population and lack of life skills. It further helps in enhancing the abilities of people who are physically challenged. The Inherent Body Sensor network was brought into existence to boost the mechanism of recovery. Further, it has been found from the ontology-based automation architecture that IoT will help in managing information very quickly. [23] The Early Childhood Education program, the Intelligent City medical recovery program, and the Integrated Goal

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Technology program are just a few of the numerous services that may be attributed to the Internet of Things. [22]

C. Health Solutions Using Smartphone:

It was noticed that the smartphone's sensor-equipped electronic device control system was this far along. In the healthcare industry, specific mobile applications are offered to assist patients, offer medical education, and offer preparatory training. Numerous applications and hardware items that portray smartphones as helpful medical tools are now in development. Table I [23]. displays a list of the top popular mobile apps for general health. [22]

TABLE I. SMARTPHONE APPLICATIONS FOR HEALTHCARE

Apps	Description
Health Assistant	This person monitors vital signs such as body temperature, blood pressure, weight, and other physical activity.
Healthy Children	Assists in locating a pediatrician in the area and makes a prompt request.
Google Fit	This wearable tracks your cycling, walking, and running activities automatically using sensors.
Noom Walk	This app tracks users' steps and is similar to a pedometer for health and fitness.
Heart Rate Monitor	With the help of this app, users can measure, track, and record their heart rates in real time for further study.
Eye Care Plus	By evaluating eye health data, this software aids in the natural improvement of eyesight.
Blood Pressure Watch	This wearable gadget measures, records, and gathers blood pressure information.

The collection and transmission of sensitive patient health data through these systems can raise concerns about privacy and security. The use of internet and wireless connectivity for data transmission adds to these apprehensions. Network or device issues could disrupt real-time monitoring. Potential for technology overload on patients. Manual health tracking may be preferable for some. [19] The accuracy and reliability of readings from wearable/home-use medical devices need to be validated by clinicians before clinical use. The initial setup and operation of such systems may require technical knowledge and support that many users lack. [20] Data integration encounters difficulties when trying to achieve interoperability between systems provided by different vendors. The ongoing costs of cloud infrastructure, technical support, and maintenance may not be affordable for everyone.

To enhance the performance of IoT-based health monitoring systems, several strategies can be implemented. Using edge computing to optimize data processing is one important strategy. This reduces latency and bandwidth utilization by processing data closer to the source. The reaction times and overall system performance of edge devices can be greatly enhanced by implementing lightweight algorithms. Moreover, concentrating on energy economy through the use of low-power CPUs and communication protocols contributes to energy conservation and longer battery life for devices. Adaptive sample rates and sensor duty cycling are two examples of network optimization strategies that can save power consumption and extend the life of monitoring equipment. System performance can be further improved by guaranteeing real-time data transfer through dependable communication channels and streamlining communication protocols for high data throughput.

Multiple solutions are needed to achieve high accuracy and dependability in an IoT-based health monitoring system. A strong system architecture with failover methods, redundancy for important components, and stringent testing procedures are necessary to ensure reliability. Sufficient, medically approved sensors are necessary for precise data gathering, which is necessary for efficient patient monitoring. Sophisticated data analytics and machine learning algorithms can improve precision by eliminating noise and detecting patterns of abnormalities in health. Accuracy is further ensured by cross-referencing current data with historical medical records and validating data in real time. In addition, low latency and continuous data transmission depend on a reliable and secure network infrastructure. Reliability and accuracy must be sustained over time by regular calibration and maintenance of sensors and equipment.

III. CONCLUSION

This review study has provided a comprehensive analysis of IoT-based health monitoring systems, emphasizing their use, function in healthcare, reliability, and performance enhancements. An innovative strategy for improving patient care and monitoring capabilities is the integration of IoT technologies in healthcare environment. An overview of recent developments in fitness-tracking applications and their advantageous effects on existing technologies are given in this study. After reviewing these studies, we learned about the specific approach taken by inventors to create these systems utilizing sensors such as ambient temperature sensors, Arduino microcontroller boards, local servers, or cloud computing. Based on a thorough analysis of these studies, Amazon Web is the ideal cloud for this type of application because of its strong security features and approachable design. Alternative cloud services can be utilized for the transmission of data and routine monitoring, an approach focusing on specific diseases or age groups can enhance the effectiveness of these tracking systems. The gathered data can be utilized, through Machine Learning techniques, to enable individuals to predict their fitness levels based on personalized factors like Body temperature, Glucose levels Blood Pressure, and other metrics, as these vary among individuals.

Future research projects should investigate the more advanced uses of IoT in healthcare, such as integrating IoT with emerging technologies like artificial intelligence for predictive analytics, facilitating interoperability across various platforms, and tackling cybersecurity issues to protect patient data. Working together across interdisciplinary disciplines will be essential to realizing IoT's full potential and transforming healthcare delivery worldwide.

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